



An experimental and modeling study of ammonia pyrolysis

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ABSTRACT

Ammonia (NH₃) is a promising carbon-free fuel and a hydrogen carrier. In recent years, there has been a large number of experimental and numerical studies to understand the chemical kinetics of NH₃ in moderate to extremely complex systems. This study focused on understanding the chemical kinetics of NH₃ in a simple system (pyrolysis). Shock-tube experiments were performed to monitor the NH₃ time history profiles during pyrolysis, with and without the presence of H₂, using a new laser absorption diagnostic near 10.4 μm. The pyrolysis experiments were conducted for mixtures of ~ 0.5% NH₃/Ar and ~ 0.42% NH₃/2% H₂/Ar behind reflected shock waves, near atmospheric pressure, and over a temperature range of 2100–3000 K. Using the data from the present study as a guide, along with NH₃ pyrolysis data from the literature, a detailed chemical kinetics mechanism for NH₃ pyrolysis is proposed herein. This mechanism was assembled using available reaction rate constants from the literature. The mechanism showed excellent agreement with the experimental results, as well as with the literature data. Additionally, an assessment of 15 detailed NH₃ chemical kinetics mechanisms on their capabilities of predicting the new pyrolysis experiments was performed. The assessment showed that these literature mechanisms yield significantly different predictions, with only one model producing acceptable results for the majority of the NH₃ pyrolysis experiments. The effect of pyrolysis reactions on the prediction of oxidation data was investigated by updating the pyrolysis sub-mechanism of selected literature models with the reactions of the present pyrolysis model. The prediction of the updated literature models significantly improved for ignition delay time and flame speed literature data, indicating the importance of pyrolysis reactions for high-temperature oxidation chemistry.

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1. Introduction

Most of the electricity currently produced worldwide is coming from fossil fuel combustion, which increases the level of CO₂ in the atmosphere and contributes to global warming. While large improvements in terms of CO₂ emissions have been made by progressively transitioning power plants from coal to natural gas (NG), the ever-growing demand for energy makes it desirable (and anticipated in many areas of the world) that most of the increase in electricity production will be covered by renewable energies.

However, most of the renewable sources of energy (solar, wind, tidal, etc.) are typically intermittent, which introduces a need for energy storage. While the storage of energy can be done under various forms such as batteries for direct electricity storage or pumped-storage hydropower to produce electricity on demand, only chemical energy storage appears to be both effective and affordable at grid-scale levels [1,2]. Chemical energy storage can be made in the form of hydrogen (H₂) production by water electroly-

sis using the electricity produced in excess during the production time of these renewable sources.

It is also worth mentioning that since a large amount of hydrogen would need to be produced, at an affordable cost, another (cheaper) way of producing H₂ without carbon emissions would be to use fossil fuels as a source but coupled with carbon capture. Such types of H₂ production are thought to still be net-negative in terms of carbon emissions and have the advantage of possibly using various feedstocks such as biomass or even waste plastic.

Regardless of the way it is produced, hydrogen is an extremely flammable gas, and its economical storage and transport remain important, unsolved challenges [1,2]. These processes are also unsafe due to the very high flammability and diffusivity of H₂. A practical solution to these drawbacks can be found by converting H₂ into a carbon-free, hydrogen-carrying fuel such as ammonia (NH₃) via a well-established industrial process [2]. Ammonia has a very low flammability, is difficult to ignite, and can be transported and stored in relatively low-pressure tanks, similar to propane [1]. These facts explain why ammonia has been considered as a hydrogen-carrying energy vector by many countries [1,2].

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While producing hydrogen back from ammonia is possible, such an additional step would increase the cost and the energy spent during the whole process. The option of using ammonia directly as a fuel is thus being considered. However, the direct utilization of NH_3 as a fuel remains challenging due to its low flammability. For example, the low reactivity of ammonia is problematic for gas turbines as it necessitates increasing the residence time by reducing the air flow [1,3]. This workaround reduces the level of turbulence and therefore the effectiveness of the mixing and the efficiency of the combustion [4]. When compared to hydrocarbons, NH_3 necessitates high temperatures to ignite in laboratory devices [5–7], which translates into a high resistance to ignition in compression-ignition engines [8,9]. A recent review on the use of NH_3 in this type of engine [10] concluded that the implementation of ammonia in a single-fuel concept is challenging due to the extremely high compression ratios required. The aforementioned low burning velocity of NH_3 , observed many times when measuring laminar flame speed in the laboratory [11–20], is also not advantageous for spark-ignition engines, as it can induce large amounts of unburned fuel, combustion instabilities, and, potentially, knock. A few strategies have been tried to mitigate these drawbacks from NH_3 , such as very advanced injection timing [21] or various ignition methods [22]. However, as mentioned by Mounaïm-Rousselle and Brequigny [23], a surprisingly small number of studies have been performed for NH_3 as a (single) fuel in engines.

It is also worth mentioning that using ammonia as a single fuel can lead to significant NO_x emissions [1,2,24], which are regulated pollutants. However, post-treatment solutions or two-stage combustion [25] are possible for fixed apparatuses like gas turbines, and NO_x emissions similar or lower than with regular gasoline are observed for stoichiometric or slightly rich fuel/air mixtures in engines, with the fuel-lean cases being problematic [26]. Finally, it has been shown that a blend of hydrogen and ammonia possibly reduces NO_x emissions to half of those from gasoline engines [27].

Thus, it is often considered that NH_3 should be mixed with fuels such as H_2 [28–34], syngas [30], (a mixture mostly composed of CO and H_2), or natural gas (which can be biogas) [28,35–38]. However, while these studies with binary mixtures are of course necessary and important to facilitate the integration of ammonia into the energy mix, it must be kept in mind that the combustion chemistry of ammonia itself is still not very well known, as illustrated by the flurry of experimental and numerical studies carried out recently [5,39–44]. More specifically, a review of the recent literature showed that while ignition delay time measurements are relatively well predicted by modern mechanisms, the same mechanisms do not provide acceptable predictions for any of the species measured in speciation studies [44]. Stagni et al. [43] concluded that “a comprehensive understanding of ammonia’s kinetic behavior is still an open challenge, especially below 1200 K and under diluted conditions”. This conclusion from Stagni et al. was also reached by a very recent study in a Jet Stirred Flow Reactor by Manna et al. [45], where literature models failed to reproduce data for the oxidation of NH_3 . Interestingly, pyrolysis data were particularly poorly predicted in this study, indicating a need to revisit NH_3 pyrolysis too. A recent study on ammonia combustion from Glarborg et al. [42] showed that the largest uncertainties are present under low-temperature conditions, especially with high O_2 concentrations, and that the experimental data under these operating conditions in support of proper model development are still very limited. Finally, note that most models also fail to predict accurately the laminar flame speed of pure ammonia [46].

In the authors’ opinion, there are few if any other molecules for which such a large number of detailed kinetics models have been proposed within the past 5–6 years; picking what seems to be the best detailed kinetics model to design a combustion device running on NH_3 could be a difficult task to start with. There is

therefore a need for assessing the best models available in the literature and for proposing new experiments and further validated models. This need for further model assessment can be illustrated by the results in Fig. 1, where a profile of NH_3 decomposition obtained during this study, for a simplistic NH_3/Ar mixture, is compared to several detailed kinetics mechanisms from the literature [5,16,34,40–43, 47–54]. The details behind the experimental measurement in Fig. 1 are provided in the subsequent section but one can see that the vast majority of the models cannot be considered as accurate, even for such a simple system.

The aims of this paper were threefold. First, new experiments were conducted for NH_3 pyrolysis by following the evolution of the NH_3 concentration profile with time via a laser absorption setup in a shock tube using mixtures of NH_3 and NH_3/H_2 diluted in Ar. The second objective of this study was then to compare detailed kinetics models from the literature with the new data to assess the models’ accuracy. The final objective of this study consisted of proposing a new detailed kinetics mechanism capable of accurately reproducing the present data as well as literature data for NH_3 pyrolysis. To the best of the authors’ knowledge, this study presents the first shock-tube quantitative laser absorption speciation measurements of NH_3 at high temperatures. This paper starts by presenting the experimental apparatus and laser absorption diagnostics, followed by a description of the developed NH_3 pyrolysis mechanism. The experimental results and numerical predictions of the new model are presented next, followed by a comparison between the new model predictions and other literature data of NH_3 pyrolysis. Finally, a comparison of the experimental data to the numerical predictions from the detailed chemical kinetics models of NH_3 from the literature is presented.

2. Experimental methods

2.1. Facilities

Experiments to measure the NH_3 speciation profiles were conducted in a stainless-steel shock tube consisting of a 6.78-m long driven section and a 3.0-m long driver section with internal diameters of 16.2 cm and 7.62 cm, respectively. Polycarbonate diaphragms with a thickness of 0.254 mm were used to separate the two sections. The driven section was vacuumed to a pressure of $\sim 10^{-5}$ Torr prior to each experiment using a combination of a mechanical pump and a turbomolecular pump. Five piezoelectric pressure transducers, mounted along the last 2 m of the driven section, were used to monitor the propagation of the incident shock wave, thus allowing the determination of the incident-shock velocities at four locations. These velocities are then used to extrapolate the incident-shock velocity (v_s) to the endwall location. v_s is then used in conjunction with the initial driven-section gas temperature (T_1) and pressure (P_1) to calculate the temperature (T_5) and pressure (P_5) behind the reflected shock wave (RSW). The estimated uncertainty in T_5 and P_5 are 0.8% and 1.0%, respectively [55].

Mixtures were prepared in a separate stainless-steel tank via the partial pressure method. The tank was vacuumed overnight prior to mixture preparation to reach a pressure of $\sim 10^{-6}$ Torr. Three pressure measurement devices were used during mixture preparation with full-scale ranges of 0–1.333 kPa, 0–133.3 kPa (both MKS 626), and 0–1725 kPa (Setra 225). All gasses were supplied by Praxair with a purity level of 99.995% for NH_3 , and 99.999% for H_2 and Ar. Before performing experiments, the mixtures were allowed to mix for at least 1 hour. Two highly diluted mixtures, with and without the presence of H_2 , were utilized to study the kinetics of NH_3 in the shock tube. The two mixtures were initially prepared as 0.7% $\text{NH}_3/99.3\%$ Ar and 0.7% $\text{NH}_3/2\%$ $\text{H}_2/97.3\%$ Ar. Although the same respective mixtures were used to fill the driven section prior to each experiment, the mole fraction

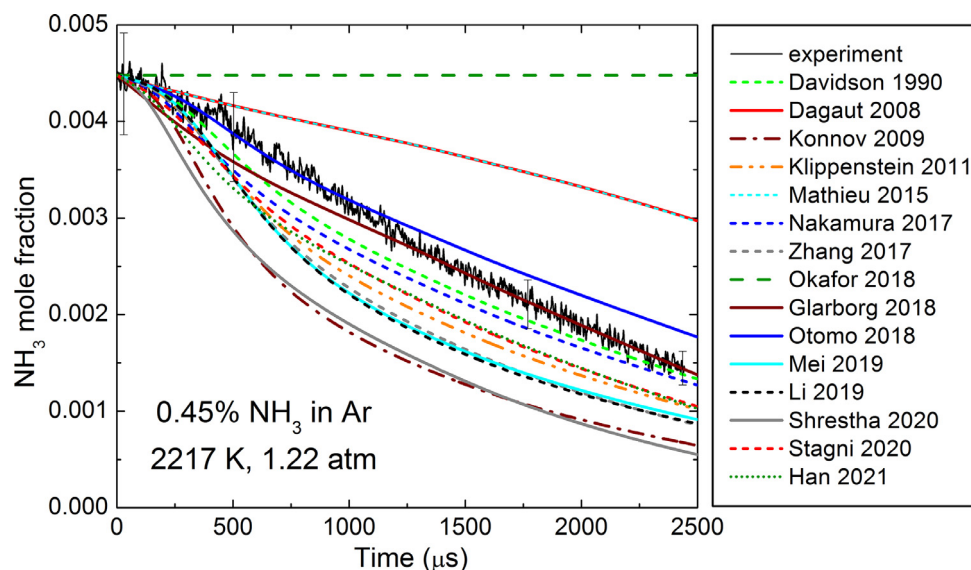


Fig. 1. Comparison of experimental NH_3 speciation profile from this study and available kinetics models from the literature. Conditions: 0.45% NH_3 in Ar at 2217 K, and 1.22 atm.

of NH_3 in the test section was found to vary from one experiment to the other. This variation is due to the strong tendency of NH_3 to adsorb onto the stainless-steel walls of the whole apparatus (mixing tank, manifold, and shock tube). The variation in the initial NH_3 mole fraction between each experiment could be attributed to parameters such as the temperature of the shock tube and time between the filling of the driven section and the experiment. Thus, the NH_3 mole fraction in the mixture was measured individually for each experiment using the laser absorption signal prior to the arrival of the incident shock wave (discussed later). To the authors' knowledge, the direct measurement of the NH_3 mole fraction present in the mixture using a quantitative laser absorption method is performed herein for the first time in a kinetics study. A discussion illustrating the importance of the direct measurement of NH_3 is provided in section S3 of the supplementary material (SM). Note that most of the previous experimental studies of NH_3 used a passivation method to reduce the effect of apparatus wall adsorption on the variation in the initial NH_3 mole fraction in the mixture, yet, this passivation method is far from being perfect, as can be seen from the discussion of the passivation tests provided in section S4 of the SM.

2.2. Laser absorption diagnostics

Measurements of NH_3 speciation profiles were obtained using a tunable quantum cascade laser (QCL) that generates narrow-linewidth light near $10.4 \mu\text{m}$. The QCL was tuned to 957.839 cm^{-1} to access the $\text{sQ}(12,12)$ transition in the ν_2 fundamental band of NH_3 —a transition that has been spectroscopically characterized in our previous work [56]. The generated light was split into incident (I_0) and transmitted (I_t) beams using a ZnSn flat beamsplitter. Two BaF_2 windows, located 1.6 cm from the endwall, were used to pass I_t through the shock tube. The intensities of the beams were monitored via two, 1-MHz bandwidth, HgCdTe photovoltaic detectors (InfraRed Associates). Bandpass filters centered on 10,400 nm (FWHM 100 nm) and 10,177 nm (FWHM 708 nm) were used to cover the detectors receiving I_t and I_0 , respectively. A flip mirror was used to send the laser beam to a wavemeter (Bristol 671) to regularly monitor the frequency of the light.

Figure 2 presents a representative experimental result showing a laser transmission trace (Fig. 2a) and the corresponding pressure during the experiment (Fig. 2b) measured via the last piezoelec-

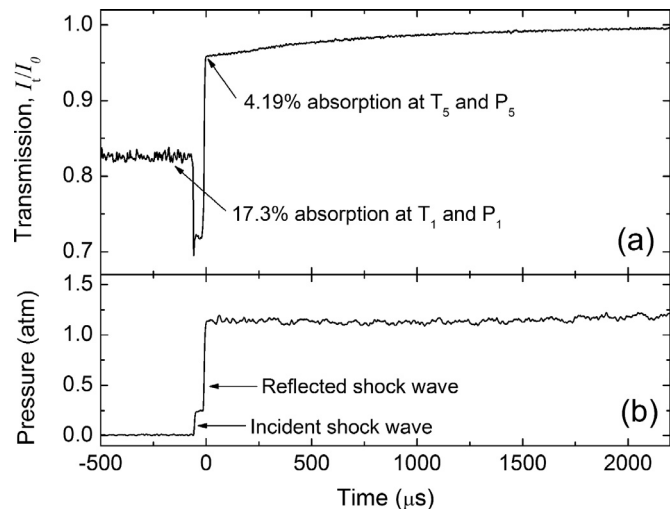


Fig. 2. Representative experiment showing (a) laser transmission and (b) sidewall pressure traces acquired in a mixture of 0.439% NH_3/Ar at 2373 K, 1.16 atm.

tric pressure transducer located on the same plane as the BaF_2 windows. In the present study, the relative experimental transmission signal (I_t/I_0) was used to (i) determine the initial NH_3 mole fraction present in the mixture after filling the test section (i.e. at T_1 and P_1), and (ii) to follow the NH_3 time history profile at the high-temperature test conditions. Both tasks invoke the Beer-Lambert law, $I_t/I_0 = \exp[-k_\nu(t) \cdot X_{\text{NH}_3}(t) \cdot P(t) \cdot L]$, to calculate the time-varying NH_3 mole fraction ($X_{\text{NH}_3}(t)$) given that the time-varying absorption coefficient ($k_\nu(t)$), the time-varying total mixture pressure ($P(t)$), and the path length (L) are all known. $P(t)$ was experimentally measured and the diameter of the shock tube was used as L , while the evaluation of $k_\nu(t)$ is discussed later. The absorption coefficient at room-temperature ($k_{\nu,296\text{K}}$) can be calculated using the known line strength and line shape of the $\text{sQ}(12,12)$ transition at 296 K provided in Alturaifi and Petersen [56]. Since NH_3 strongly adsorbs to stainless-steel walls, the initial NH_3 mole fraction ($X_{\text{NH}_3,0}$) in the mixture varied after filling the test section, as mentioned before. Thus, $X_{\text{NH}_3,0}$ was measured directly for each experiment using T_1 , P_1 , the known value of $k_{\nu,296\text{K}}$ from [56], and

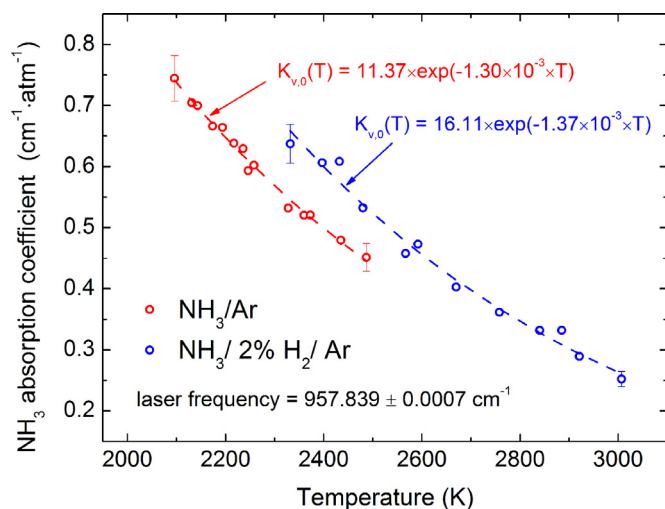


Fig. 3. Spectral absorption coefficient values of NH_3 immediately behind the reflected shock wave, $k_{\nu,0}$, as a function of T_5 . Dashed lines present the best-fit of the experiments for a mixture of $\sim 0.5\% \text{NH}_3$ in Ar (red) and $\sim 0.42\% \text{NH}_3/2\% \text{H}_2$ in Ar (blue). The pressure varied between 0.88 and 1.26 atm.

the average transmission signal prior to the arrival of the incident shock wave (e.g. the signal from $-500 \mu\text{s}$ to $-50 \mu\text{s}$ in Fig. 2a).

Subsequently, $X_{\text{NH}_3,0}$ is used to calculate the initial absorption coefficient at high temperature ($k_{\nu,0}$) using the transmission signal immediately after the passage of the RSW (e.g. the signal at 4.19% absorption in Fig. 2a) in conjunction with the known T_5 and P_5 . Using the values of $k_{\nu,0}$, P_5 , and L , $X_{\text{NH}_3}(t)$ can be calculated via the Beer-Lambert law using the remaining I_t/I_0 signal in the experiment. A small correction for $X_{\text{NH}_3}(t)$ was performed due to the small pressure variation during the experiment (discussed later). The uncertainty in the calculated mole fraction is conversely estimated to be $\pm 12\%$. This uncertainty was calculated for selected experiments using the root-sum-squares method of the major uncertainty sources. Each source was perturbed within its uncertainty to assess its effect on the calculated mole fraction. The dominant uncertainty source in this study is the uncertainty in determining the initial NH_3 mole fraction present in the mixture. Several factors contribute to this uncertainty such as uncertainties in the spectroscopic parameters, T_1 , and P_1 , but the dominant factor is the uncertainty in the laser frequency relative to the $\text{SQ}(12,12)$ transition's line center. It should be noted that this dominant factor is important in this study due to the narrow line width (typically $\sim \pm 0.002 \text{ cm}^{-1}$ FWHM) of the $\text{SQ}(12,12)$ transition at the initial low pressures of the driven section (i.e. at P_1 , typically $\sim 8 - 20$ torr).

The extracted $k_{\nu,0}$ from all the experiments are plotted against T_5 in Fig. 3. The uncertainty in $k_{\nu,0}$ is estimated to be $\pm 7\%$ mainly due to the initial NH_3 mole fraction uncertainty. The best-fits of the $k_{\nu,0}$ data yield the absorption coefficient temperature dependency ($k_{\nu,0}(T)$) for the two mixtures investigated in the present study. During the experiment test time, the temperature is changing slightly due to the non-ideal pressure variation (discussed in the next paragraph). This small temperature change alters k_{ν} during the test time, and the relation of $k_{\nu,0}(T)$ is used to calculate the time-varying absorption coefficient $k_{\nu}(t)$.

Figure 4 shows the analyzed NH_3 speciation data for the same experiment shown in Fig. 2. The pressure during the test time exhibits a small, non-ideal pressure variation (typically $dP/dt < 3\%$) which, over time, alters the conditions behind the RSW, compared to the initial calculated T_5 and P_5 . This pressure (and temperature) variation also alters the calculated $k_{\nu,0}$ values during the test duration. Thus, a time-varying temperature profile ($T(t)$) was calculated based on the time-varying pressure profile $P(t)$ obtained

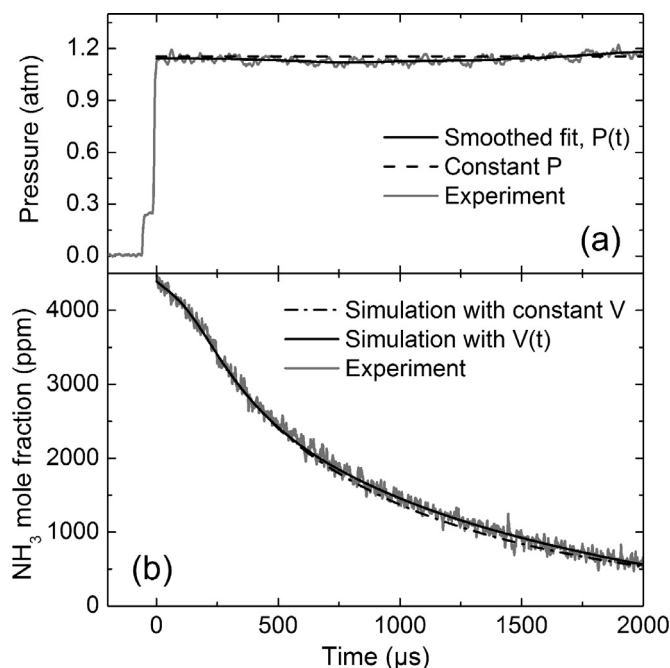


Fig. 4. Representative analyzed experiment and numerical modeling result for the same experiment conditions in Fig. 2. (a) Pressure profile and extracted smoothed fit $P(t)$. (b) Experimental NH_3 speciation profile and simulation using the present mechanism using either constant V or $V(t)$.

using a smooth fit of the experimental pressure signal (shown in Fig. 4a). This $T(t)$ was calculated by assuming isentropic expansion/compression during the test time; an assumption that was validated previously for similar conditions [57]. Subsequently, the time-varying absorption coefficient profile, $k_{\nu}(t)$, was calculated using the best-fit expressions of $k_{\nu}(T)$ (shown in Fig. 3) in conjunction with the calculated $T(t)$. This correction of k_{ν} had a very small effect (e.g. $\sim$ maximum of 30 ppm for the experiment shown in Fig. 4b) on the calculated $X_{\text{NH}_3}(t)$ but was nonetheless performed for each experiment. More information on the pressure variation analysis can be found in Mulvihill et al. [58], while additional information on the laser absorption method and post-processing procedures can be found in Alturaifi et al. [59].

2.3. Modeling of the shock-tube data

All shock-tube modeling predictions were conducted using AN-SYS Chemkin software via a closed, homogeneous batch reactor model assuming constant internal energy and employing the volume-as-function-of-time (VTIM) method [60]. The non-ideal pressure variation during the experiments had a small effect on the modeling prediction of the NH_3 speciation profiles and was accounted for when performing the numerical simulations. Accounting for the non-ideal pressure variation was conducted by implementing a time-varying volume profile $V(t)$ derived from the experimental $P(t)$ profile by assuming an isentropic expansion/compression. The modeling predictions of NH_3 using the present mechanism (discussed next) with constant V and with $V(t)$ are shown in Fig. 4b to highlight the small effect of including the pressure variation when modeling the data herein. Modeling the data of the present study using Chemkin can be performed by specifying the pressure profile as the extracted smooth-fit experimental pressure profiles, which are included in the SM. More information about modeling shock-tube speciation data can be found elsewhere [59,60].

3. Chemical kinetics model

To the knowledge of the authors, the Davidson et al. [47] mechanism was the only model specifically assembled for NH_3 pyrolysis and was only validated against experimental data from the same study. The present study used the Davidson et al. mechanism as a base model with four main stages of improvements. The first stage of improvement was to add a few important NH_2 , NH , and N_2H_2 reactions that were not present in the Davidson et al. mechanism. The second stage was to add reactions involving N_2H_3 , N_2H_4 , and H_2NN for the N_2H_x sub-mechanism. The selection of the added reactions in the first and second stage was guided by the pyrolysis sub-mechanism of several modern detailed kinetics models of NH_3 . These reaction additions effectively led to an overall mechanism that is generally consistent with the modern NH_3 mechanisms. The third was to adopt the thermochemical data reported by Bugler et al. [61] where they performed high-accuracy *ab initio* calculations to obtain accurate and consistent thermochemical parameters for nitrogen-containing species. Finally, the fourth stage was to update individual reaction rates to best fit the experimental data reported herein, as well as the few pyrolysis data available in the literature. A survey of the reported reaction rates in the literature was conducted, and multiple optimization iterations were performed to evaluate the effect of rate constant selection on the numerical prediction of the experimental data. All the selected reaction rate constants were taken directly as reported in the literature, and no tuning of individual reaction rates was performed. The proposed mechanism significantly improved the predictions of the NH_3 speciation profiles provided in the present study and was validated against the experimental data reported in the Davidson et al. study, and the few other sets of pyrolysis data reported in the literature [45,62,63]. The proposed NH_3 pyrolysis mechanism, which is described in Table 1, contains 13 species and 48 reactions, and the mechanism and thermodynamic data in Chemkin input format are provided in the SM.

The model analysis showed that R4–5 are the main NH_3 -consuming reactions which were both included in the Davidson et al. mechanism. The reaction rate of $\text{NH}_3 + \text{M} \rightleftharpoons \text{NH}_2 + \text{H} + \text{M}$ (R4) was directly adopted from the study of Davidson et al., who reanalyzed experimental data from four earlier studies to obtain a rate constant with lower uncertainty. Recently, Altinay et al. [77] measured the rate constant of R4 (in reverse) at low temperature using pulsed-laser photolysis of NH_3 . However, adopting the rate constant reported by Altinay et al. in the present mechanism significantly accelerated the NH_3 consumption compared with the experimental profiles. It should be noted that Altinay et al. reported the rate of R4 with N_2 as a third-body collision partner, while the present data were obtained in highly diluted Ar mixtures. Thus, a collision efficiency of Ar coupled with the Altinay et al. rate constant of R4 might improve the prediction of the present experiments. On the other hand, the Davidson et al. mechanism employed a rate constant of $\text{NH}_3 + \text{H} \rightleftharpoons \text{NH}_2 + \text{H}_2$ (R5) from the Michael et al. [78] flash photolysis shock-tube study. The present study adopted an updated rate constant for R5 from the study of Friedrichs and Wagner [66], where they directly measured the reverse rate $\text{NH}_2 + \text{H}_2 \rightleftharpoons \text{NH}_3 + \text{H}$ (R5) using frequency modulation spectroscopy in a shock tube.

R6–13 are reactions of NH_2 with M, H, N, NH, and NH_2 . Davidson et al. [47] measured the reaction rate of $\text{NH}_2 + \text{H} \rightleftharpoons \text{NH} + \text{H}_2$ (R7) to be $4.0 \times 10^{13} \exp(-3650/RT)$ in cm, mol, and s, with an uncertainty of 50%. However, the present mechanism utilized the reaction rate proposed by Miller and Bowman [68] for R7 (which is $\sim 70\%$ higher than that measured by Davidson et al.). Although the selection of R7 had a minimal effect on the experimental NH_3 consumption profiles, selecting the rate proposed by Miller and Bowman considerably improved the prediction of the experimen-

tal NH_2 and NH peak data used during model validation (discussed later). All the reaction rates of NH_2 with NH or NH_2 were directly adopted from the theoretical calculations by Klippenstein et al. [70] except $\text{NH}_2 + \text{NH} \rightleftharpoons \text{N}_2\text{H}_2 + \text{H}$ (R10) where the rate coefficient provided by Davidson et al. [47] was adopted. R10 is the major pathway to form N_2H_2 which plays a critical role in controlling the NH_3 consumption (as shown later in the chemical analysis). Selecting the rate of R10 proposed by Klippenstein et al. accelerated the NH_3 consumption, while the Davidson et al. rate of R10 improved the prediction of the experimental data.

R14–18 are reactions of NH with M, H, N, and NH . The present mechanism adopted the rate constant for the decomposition reaction of NH (and NH_2) measured by Deppe et al. [67]. The reaction $\text{NH} + \text{H} \rightleftharpoons \text{H}_2 + \text{N}$ (R15) is the most-promoting reaction of NH_3 during pyrolysis at high temperature. R15 reported rate constants in the literature are limited, and the present study adopted the rate constant proposed by Miller and Bowman [68]. R19–22 are reactions of NNH with M, H, NH , and NH_2 . The rate constants for all these NNH reactions were adopted from the Miller and Bowman study except the dissociation reaction $\text{NNH} + \text{M} \rightleftharpoons \text{N}_2 + \text{H} + \text{M}$ (R19a–b). The present mechanism models this dissociation reaction per the recommendation of Nakamura et al. [40] who used the rate constant reported in Dean and Bozzelli [71].

R23–27 are N_2H_2 dissociation and isomerization reactions and reactions with H, NH , and NH_2 . The rate constants of the N_2H_2 dissociation and isomerization reactions to form $\text{NNH} + \text{H}$ (R23) and H_2NN (R24) were taken from Dean and Bozzelli [71]. Linder et al. [72] performed *ab initio* computation to determine the rate constant of N_2H_2 reactions with H and NH_2 (R25 and R27). Chuang and Truhlar [79] also performed theoretical computations to determine an improved rate constant of R25 by accounting for multi-dimensional tunneling. The present study chose the rate constants determined by Linder et al. due to the significant deviation in numerical prediction of the NH_3 consumption rate (compared with the experimental measurements) when adopting the rate constant determined by Chuang and Truhlar. It should be noted that R25 is an important promoting reaction of NH_3 (discussed later).

R28–36 and R37–43 are reactions of N_2H_3 and N_2H_4 , respectively. The major sources of the rate constant of these reactions were the review of Dean and Bozzelli [71] and the study of Konnov and De Ruyck [73]. Additionally, the rate constant selection was guided by the study of Nakamura et al. [40] which validated the N_2H_x sub-mechanism in weak flames where this sub-mechanism chemistry is important. An exception is the hydrazine decomposition reaction $\text{N}_2\text{H}_4 + \text{M} \rightleftharpoons \text{NH}_2 + \text{NH}_2 + \text{M}$ (R37 in reverse) where the rate constant from Klippenstein et al. [70] was adopted due to its superior prediction of the N_2H_4 data used in the mechanism validation (discussed later). Finally, H_2NN is a major product of the $\text{NH}_2 + \text{NH}_2$ reaction (R13) at low temperature, and its reactions (R44–47) were adopted directly from the review of Dean and Bozzelli.

4. Results and discussion

A total of 26 experiments were conducted to measure the NH_3 speciation profiles during the pyrolysis of NH_3 , with and without the presence of H_2 , in highly diluted Ar mixtures and near-atmospheric pressure. The experiments without H_2 were conducted for effective mixtures of 0.42–0.52% NH_3 in Ar (after NH_3 adsorption on the walls, referred to herein as mixture 1) for a range of temperatures from 2096 to 2487 K, while the experiments with H_2 were performed for effective mixtures of 0.36–0.45% $\text{NH}_3/2\%$ H_2 in Ar (mixture 2) in the temperature range of 2332–3007 K. Additionally, from the speciation profiles, the half-life time of NH_3 at a given temperature was extracted for each experiment. The NH_3 half-life time is defined as the time from the

Table 1List of reactions in the proposed NH₃ pyrolysis mechanism. Reaction rates are given in Arrhenius expressions as $k = AT^n \exp[-E_a/RT]$. Units are cal, mol, cm, s, K.

No.	Reaction	A	n	E _a	Source
R1	H ₂ + M ⇌ H + H + M Third body efficiency $\epsilon_{H_2} = 2.5, \epsilon_{Ar} = 0$	4.58×10^{19}	-1.40	104,400	[64]
R2	H ₂ + AR ⇌ H + H + AR	5.80×10^{18}	-1.10	104,380	[64]
R3	N ₂ + M ⇌ N + N + M	1.89×10^{18}	-0.85	224,950	[65]
R4	NH ₃ + M ⇌ NH ₂ + H + M	2.20×10^{16}	0.00	93,470	[47]
R5	NH ₂ + H ₂ ⇌ NH ₃ + H	1.50×10^4	2.60	5975	[66]
R6	NH ₂ + M ⇌ NH + H + M	1.20×10^{15}	0.00	75,950	[67]
R7	NH ₂ + H ⇌ NH + H ₂	6.92×10^{13}	0.00	3650	[68]
R8	NH ₂ + N ⇌ N ₂ + H + H	7.00×10^{13}	0.00	0	[69]
R9	NH ₂ + NH ⇌ NH ₃ + N	9.57×10^3	2.46	107	[70]
R10	NH ₂ + NH ⇌ N ₂ H ₂ + H	1.50×10^{15}	-0.50	0	[47]
R11	NH ₂ + NH ₂ ⇌ NH ₃ + NH	5.64×10^0	3.53	522	[70]
R12	NH ₂ + NH ₂ ⇌ N ₂ H ₂ + H ₂	1.74×10^8	1.02	11,783	[70]
R13	NH ₂ + NH ₂ ⇌ H ₂ NN + H ₂	7.17×10^4	1.88	8802	[70]
R14	NH + M ⇌ H + N + M	1.80×10^{14}	0.00	74,760	[67]
R15	NH + H ⇌ H ₂ + N	1.00×10^{14}	0.00	0	[68]
R16	NH + N ⇌ N ₂ + H	3.00×10^{13}	0.00	0	[68]
R17	NH + NH ⇌ N ₂ + H + H	2.50×10^{13}	0.00	0	[69]
R18	NH + NH ⇌ NH ₂ + N	5.66×10^{-1}	3.88	342	[70]
R19a	NNH + M ⇌ N ₂ + H + M	1.00×10^{13}	0.50	3060	[71]
R19b	NNH ⇌ N ₂ + H	3.00×10^8	0.00	0	[71]
R20	NNH + H ⇌ N ₂ + H ₂	1.00×10^{14}	0.00	0	[68]
R21	NNH + NH ⇌ N ₂ + NH ₂	5.00×10^{13}	0.00	0	[68]
R22	NNH + NH ₂ ⇌ N ₂ + NH ₃	5.00×10^{13}	0.00	0	[68]
R23	N ₂ H ₂ + M ⇌ NNH + H + M				[71]
	0.1 atm	5.60×10^{36}	-7.75	70,340	
	1.0 atm	1.80×10^{40}	-8.41	73,320	
	10.0 atm	3.10×10^{41}	-8.42	76,102	
R24	N ₂ H ₂ ⇌ H ₂ NN				[71]
	0.1 atm	9.20×10^{38}	-9.01	67,727	
	1.0 atm	2.00×10^{41}	-9.38	68,452	
	10.0 atm	1.30×10^{45}	-10.13	70,757	
R25	N ₂ H ₂ + H ⇌ NNH + H ₂	8.50×10^4	2.63	230	[72]
R26	N ₂ H ₂ + NH ⇌ NNH + NH ₂	2.40×10^6	2.00	-1192	[71]
R27	N ₂ H ₂ + NH ₂ ⇌ NNH + NH ₃	8.80×10^2	4.05	1610	[72]
R28	N ₂ H ₃ + M ⇌ NH ₂ + NH + M	5.00×10^{16}	0.00	60,000	[73]
R29	N ₂ H ₃ + M ⇌ N ₂ H ₂ + H + M	1.00×10^{16}	0.00	37,000	[73]
R30	N ₂ H ₃ + H ⇌ N ₂ H ₂ + H ₂	2.40×10^8	1.50	0	[71]
R31	N ₂ H ₃ + H ⇌ NH + NH ₃	1.00×10^{11}	0.00	0	[73]
R32	N ₂ H ₃ + NH ⇌ N ₂ H ₂ + NH ₂	2.00×10^{13}	0.00	0	[74]
R33	N ₂ H ₃ + NH ₂ ⇌ N ₂ H ₂ + NH ₃	9.20×10^5	1.94	-1152	[71]
R34	N ₂ H ₃ + NH ₂ ⇌ H ₂ NN + NH ₃	3.00×10^{13}	0.00	0	[71]
R35	N ₂ H ₃ + N ₂ H ₃ ⇌ NH ₃ + NH ₃ + N ₂	3.00×10^{12}	0.00	0	[75]
R36	N ₂ H ₃ + N ₂ H ₃ ⇌ N ₂ H ₄ + N ₂ H ₂	1.20×10^{13}	0.00	0	[73]
R37	NH ₂ + NH ₂ + M ⇌ N ₂ H ₄ + M	5.60×10^{14}	-0.41	66	[70]
	Low-pressure limit	1.60×10^{34}	-5.49	1987	
	Troe parameters	$0.31, 1 \times 10^{-30}, 1 \times 10^{30}, 1 \times 10^{30}$			
R38	N ₂ H ₄ + M ⇌ H ₂ NN + H ₂ + M				[71]
	0.1 atm	4.00×10^{44}	-9.85	71,352	
	1.0 atm	5.30×10^{39}	-8.35	69,307	
	10.0 atm	2.50×10^{39}	-8.19	69,664	
R39	N ₂ H ₄ + H ⇌ NH ₂ + NH ₃	2.40×10^9	0.00	3100	
R40	N ₂ H ₄ + H ⇌ N ₂ H ₃ + H ₂	7.00×10^{12}	0.00	2500	[76]
R41	N ₂ H ₄ + N ⇌ N ₂ H ₃ + NH	1.00×10^{10}	1.00	2000	[73]
R42	N ₂ H ₄ + NH ⇌ NH ₂ + N ₂ H ₃	1.00×10^9	1.50	2000	[73]
R43	N ₂ H ₄ + NH ₂ ⇌ N ₂ H ₃ + NH ₃	3.90×10^{12}	0.00	1500	[68]
R44	H ₂ NN + M ⇌ NNH + H + M				[71]
	0.1 atm	5.90×10^{32}	-6.99	51,791	
	1.0 atm	9.60×10^{35}	-7.57	54,841	
	10.0 atm	5.00×10^{36}	-7.43	57,295	
R45	H ₂ NN + H ⇌ NNH + H ₂	4.80×10^8	1.50	-894	[71]
R46	H ₂ NN + H ⇌ N ₂ H ₂ + H	7.00×10^{13}	0.00	0	[71]
R47	H ₂ NN + NH ₂ ⇌ NNH + NH ₃	1.80×10^6	1.94	-1152	[71]

passage of the RSW (time zero) to the time corresponding to the consumption of 50% of the initial NH₃ mole fraction present in the mixture.

The presented experimental data can be used as validation targets for future NH₃ mechanisms. To facilitate future modeling of our experimental data, all experimental NH₃ speciation profiles and their associated pressure traces are provided in the SM. In addition, the experiment conditions and extracted NH₃ half-life time are also provided in Tables S1-S4 in the SM.

4.1. Experiment and model results

The experimental and modeling results of the NH₃ speciation profiles for mixture 1 are presented in Fig. 5. Experiments are presented for the low-, intermediate-, and high-temperature range (relative to the temperature range of the present study) on the left, middle, and right side, respectively. The modeling predictions using the proposed mechanism (shown in red lines) reproduce well the experimental measurements. The experimental profiles show

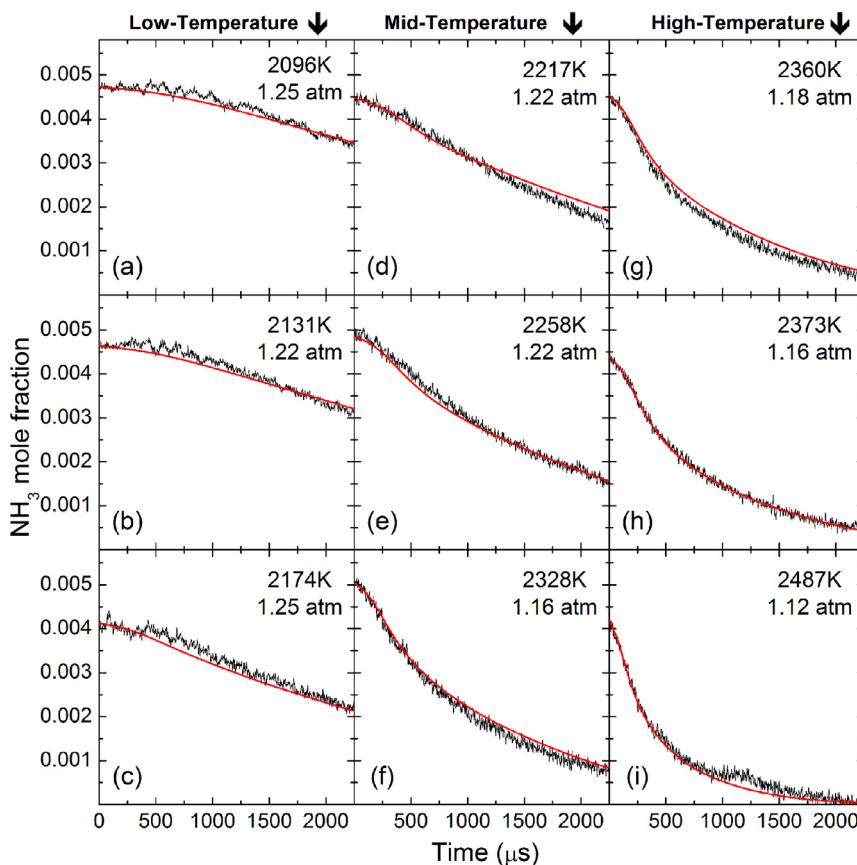


Fig. 5. Experiments and modeling results for NH_3 speciation profiles for a mixture of $\sim 0.5\%$ NH_3 in Ar.

that the consumption rate of NH_3 increases with temperature. At the lowest temperature investigated (2096 K), the NH_3 consumption starts after $\sim 1000 \mu\text{s}$, and only $\sim 25\%$ on the initial NH_3 mole fraction is consumed by the end of the test time. For the highest temperature investigated (2487 K), the initial NH_3 mole fraction is almost completely consumed after $\sim 1500 \mu\text{s}$. The NH_3 profiles displayed two-stage behavior: the first stage, which typically lasts $< 200 \mu\text{s}$, showed a small NH_3 consumption, which is followed by the second stage of faster NH_3 consumption for the remainder of the experiment. The kinetics analysis shows that the first stage is mainly sensitive to the main ammonia decomposition reaction R4 ($\text{NH}_3 + \text{M} \rightleftharpoons \text{NH}_2 + \text{H} + \text{M}$), and the best fit for this stage yielded a rate constant in agreement with the historical values. However, the determination of the rate constant of R4 with reasonable accuracy (with an uncertainty equal or better than the widely adopted rate constant of Davidson et al. [47]) was not possible mainly for two reasons: the relatively short time of the first stage decomposition, and the small amount of NH_3 mole fraction change relative to the signal-to-noise ratio of the experimental traces. The analysis of the first stage NH_3 decomposition and the estimated rate constant of R4 is available in section S2 of the SM.

Figure 6 shows the experimental and modeling results for the NH_3 speciation profiles for mixture 2 in a similar presentation to the experiments in Fig. 5. The computational predictions of the presented mechanism agree well with the experimental profiles. The experiments of mixture 2 were conducted at a higher temperature range. The addition of $2\% \text{H}_2$ to the mixture resulted in an overall faster rate of consumption of NH_3 . For example, Figs. 5f and 6a show two experiments at almost the same temperature for mixture 1 and 2, respectively. The experimental profiles show that at $1000 \mu\text{s}$, $\sim 60\%$ and 40% of the initial NH_3 mole fraction was consumed for mixtures 1 and 2, respectively. Analysis of the mech-

anism showed that the reactivity is first initiated by R4 ($\text{NH}_3 + \text{M} \rightleftharpoons \text{NH}_2 + \text{H} + \text{M}$), and then -R5 ($\text{NH}_3 + \text{H} \rightleftharpoons \text{NH}_2 + \text{H}_2$) rapidly becomes the main channel for NH_3 consumption, followed by R4. However, for mixture 2, where $2\% \text{H}_2$ is present, the rate of consumption of NH_3 via -R5 is significantly lower mainly due to the generated NH_2 from R4 reacting with H_2 and increasing the forward direction of R5, ultimately lowering the rate of the reverse direction of R5 and the total rate of NH_3 consumption.

To assess the present mechanism over the entire range of temperature, the experimental NH_3 half-life times for both mixtures are plotted against the numerical prediction in Fig. 7. The modeling was performed using a constant-pressure assumption at average conditions for each set of experiments; $0.5\% \text{NH}_3$ in Ar and 1.2 atm for mixture 1, and $0.42\% \text{NH}_3/2\% \text{H}_2$ in Ar and 1.0 atm for mixture 2. The present mechanism predicts well the NH_3 half-life times for mixture 1 across the temperature range investigated. For mixture 2, the present mechanism predicts similar NH_3 half-life times at high temperatures ($> 2700 \text{ K}$), while a small over prediction (10–20% longer times) is observed for the mid-range temperatures. At temperatures below 2500 K , the present mechanism produced NH_3 half-life times within the uncertainty of the measurements. Note that the NH_3 half-life time predictions of the mid-range temperature slightly improved when modeling each experiment with its pressure profile using the VTIM method. However, the selection of modeling the NH_3 half-life times using the average condition is to facilitate and simplify future modeling of our experiments with reasonable accuracy.

To investigate the effect of adding H_2 during the pyrolysis of NH_3 , a chemical analysis was performed with the model from the present study and summarized in Fig. 8. This figure presents the major species profiles during NH_3 pyrolysis (top), NH_3 rate of production (middle), and NH_3 sensitivity coefficient (bottom) for mix-

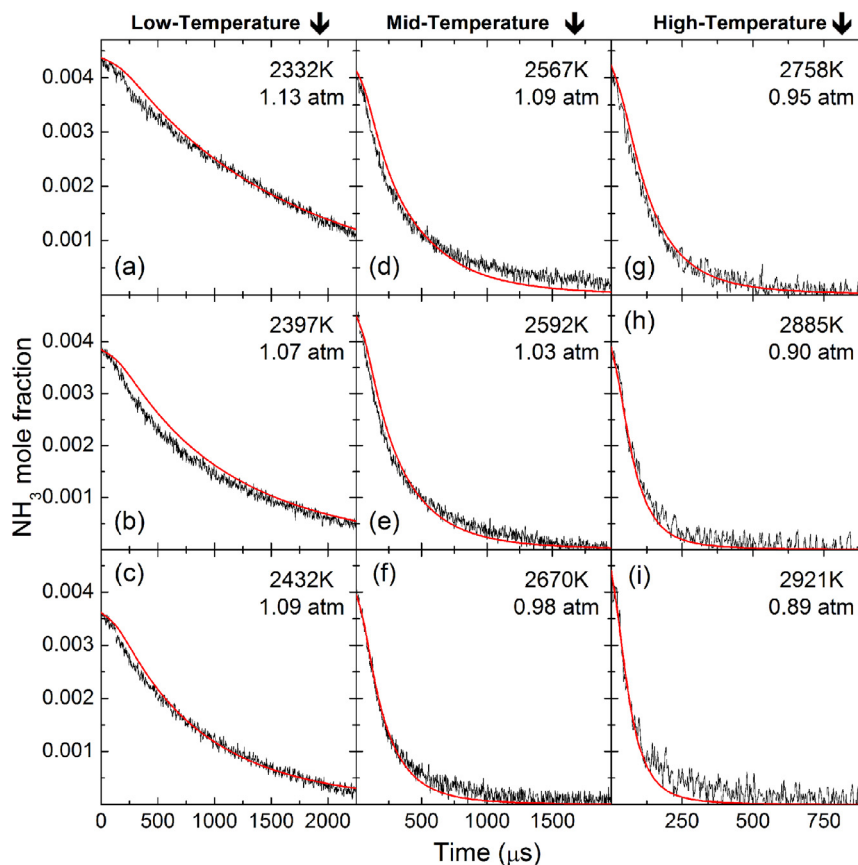


Fig. 6. Experiments and modeling results for NH_3 speciation profiles in a mixture of $\sim 0.42\% \text{NH}_3/2\% \text{H}_2$ in Ar.

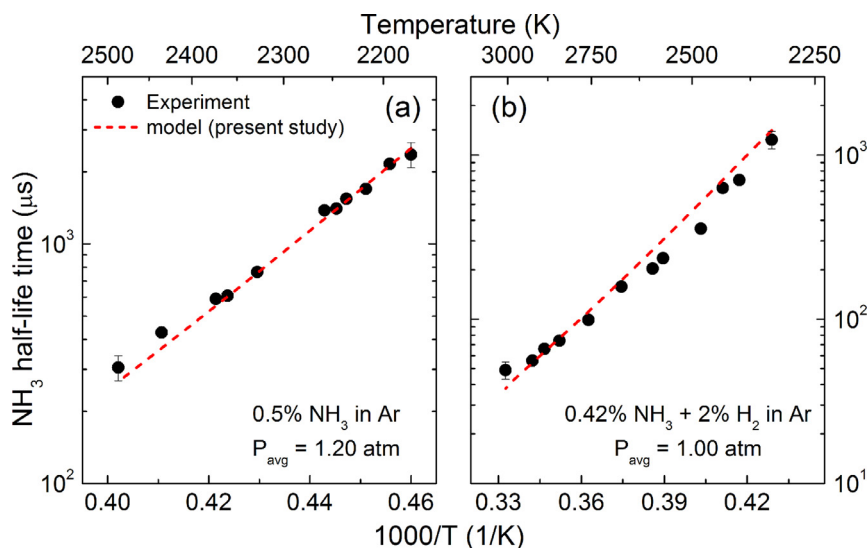


Fig. 7. Experiments and modeling results for NH_3 half-life times for a mixture of (a) $\sim 0.5\% \text{NH}_3$ in Ar, and (b) $\sim 0.42\% \text{NH}_3/2\% \text{H}_2$ in Ar.

ture 1 (left) and mixture 2 (right). The numerical computations were performed at an average condition of 2400 K and 1 atm. Focusing on the top part of the figure, the computation shows that the intermediate species NH_2 and H form immediately, followed by a rapid formation of NH , N , and N_2H_2 intermediates for both mixtures. For mixture 1, the first few microseconds show that NH_2 and H are produced in equal quantities exclusively from R4 ($\text{NH}_3 + \text{M} \rightleftharpoons \text{NH}_2 + \text{H} + \text{M}$). After a few microseconds, a significant part of the generated H -atoms from R4 is consumed via -R5 ($\text{NH}_3 + \text{H}$

$\rightleftharpoons \text{NH}_2 + \text{H}_2$), ultimately reducing the rate of production of H -atoms.

For mixture 2, NH_2 and H are produced nearly in equal quantities up to $\sim 100 \mu\text{s}$ (near NH_2 peak), with a lower concentration of NH_2 and a higher concentration of H radicals for mixture 2 compared to mixture 1. This higher level of H for mixture 2 is due to the presence of H_2 in the mixture, which enhances the forward direction of R5. After 100 μs , the larger amount of H -atoms from R4 and R5 reaches high enough levels as to promote -R5, leading

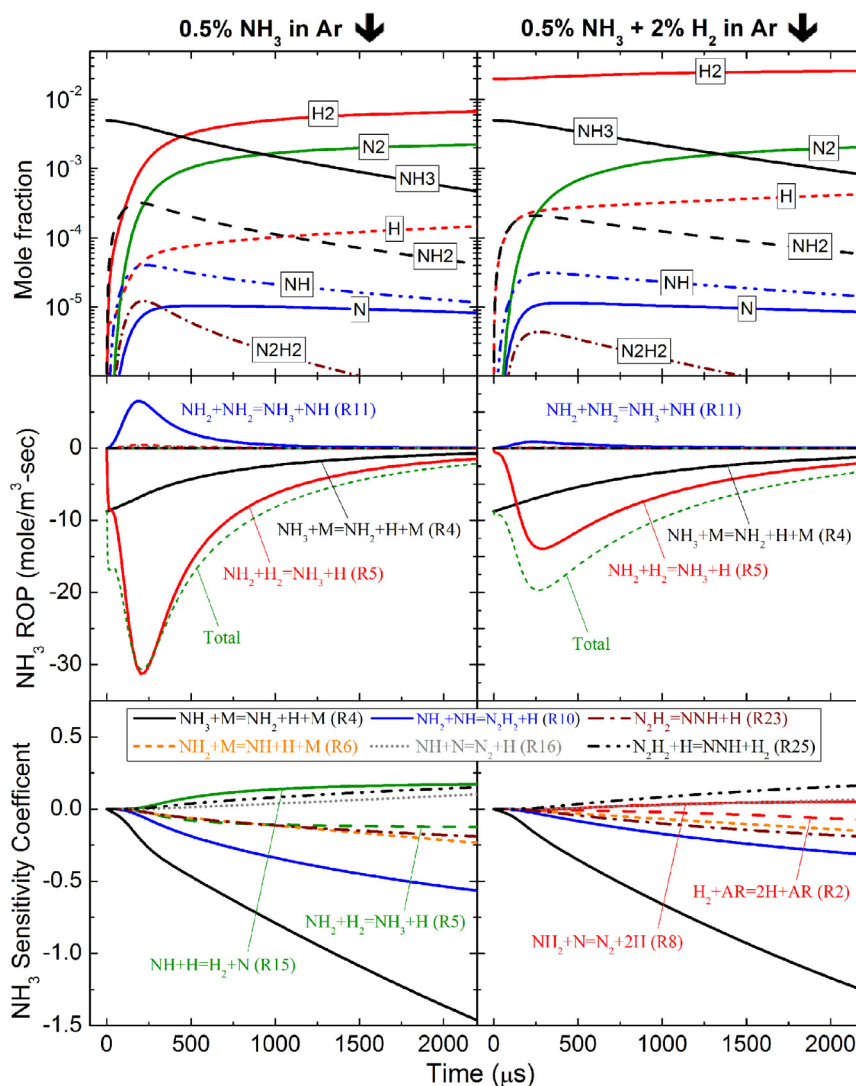


Fig. 8. Numerical computation using the present mechanism for major species profiles (top), NH_3 rate of production (middle), NH_3 sensitivity coefficient (bottom) during NH_3 pyrolysis in a mixture of 0.5% NH_3 (left) and 0.5% NH_3 /2% H_2 (right) balance Ar. Conditions: 2400 K and 1 atm.

to higher levels of NH_2 . This NH_2 is rapidly consumed by reacting with H and NH to form other intermediates NH and N_2H_2 , (via R8 and R9), and by reacting with N and NH_2 to form N_2 and NH_3 (via R10 and R11). The generated NH is consumed by reacting with H or NH to form H_2 and N_2 , respectively (via R15 and R17), in addition to its reaction with NH_2 which is the major pathway of forming N_2H_2 (via R9). Subsequently, N_2H_2 is the major pathway to form NNH by either its decomposition reaction (R23) or its reaction with H (R25). NNH is then mainly consumed via its dissociation reaction to form $\text{N}_2 + \text{H}$ (R19a-b) which is the major pathway to form N_2 . Note that although not visible in Fig. 8, NNH exists but at a level lower than 1 ppm along with H_2NN , N_2H_3 , and N_2H_4 .

The middle section of Fig. 8 shows that the rate of production for NH_3 is mainly governed by three reactions. For both mixtures, NH_3 is consumed at a steady rate by the continuous decomposition via R4 throughout the simulation time. For mixture 1, NH_3 is also rapidly consumed by reactions with H-atoms via -R5 ($\text{NH}_3 + \text{H} \rightleftharpoons \text{NH}_2 + \text{H}_2$), which rapidly becomes the major consumption pathway. On the other hand, the recombination reaction of NH_2 (R11) slows the rate of NH_3 consumption by forming $\text{NH}_3 + \text{NH}$. For mixture 2, it is visible that the consumption of NH_3 via R5 is significantly reduced compared to mixture 1. This reduced con-

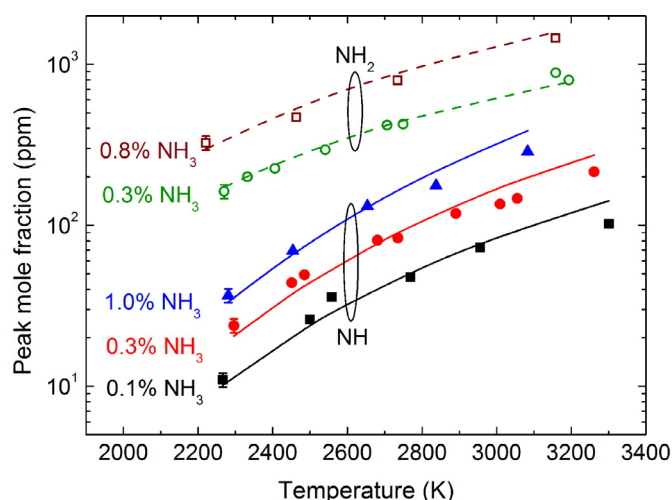
sumption is due to the large presence of H_2 in the mixture, which changes the equilibrium between NH_3 and NH_2 in R5, leading to a reduced consumption of NH_3 via -R5. As a result of this larger influence of R5, the formation of NH_3 via NH_2 recombination (R11) is also significantly reduced, and the net effect is a slower rate of NH_3 consumption for mixture 2.

NH_3 sensitivity coefficient is shown in the bottom part of Fig. 8. At early times ($< 100 \mu\text{s}$), the NH_3 decomposition reaction R4 is the only sensitive reaction for both mixtures. At later times, the generated H from R4 increases the NH_2 levels via -R5 which starts a series of secondary NH, NH_2 , and N_2H_2 reactions that influence the NH_3 profiles. For mixture 1, NH_2 reaction with NH (R10) is the second most-sensitive reaction because it is the main pathway to form N_2H_2 . N_2H_2 plays an important role in controlling the consumption rate of NH_3 with two reactions among the most-sensitive reactions; $\text{N}_2\text{H}_2 \rightleftharpoons \text{NNH} + \text{H}$ (R23) and $\text{N}_2\text{H}_2 + \text{H} \rightleftharpoons \text{NNH} + \text{H}_2$ (R25) which inhibit and promote NH_3 , respectively, due to the influence of H and H_2 on R5. Additionally, the NH reaction with H (R15) to generate H_2 is the most-promoting reaction of NH_3 as the generated H_2 reduces the reverse direction of R5. For mixture 2, R15 is not among the sensitive reactions due to the insignificance of the generated H_2 compared with the initial lev-

Table 2Experimental data used for improvement and validation of the present NH_3 pyrolysis model.

Exp. Device	Target	Conditions	Reference
Shock tube	NH_3 species profile	P: ~ 1 atm T: 2100–3100 K Mixture: $\sim 0.5\%$ NH_3 in Ar $\sim 0.42\%$ $\text{NH}_3/2\%$ H_2 in Ar	PW*
Shock tube	NH_2 and NH species profile	P: ~ 1 atm T: 2200–2800 K Mixture: 0.01–1% NH_3 in Ar	[47]
Shock tube	N_2H_4 half-life times	P: $\sim 7 - 9$ atm T: 1100–1600 K Mixture: 0.03–0.5% N_2H_4 in Ar	[62]
Flow reactor	NH_3 and H_2 species profiles	P: 1 atm T: 900–1800 K Mixture: 0.1 – 1% NH_3 in Ar or N_2	[63]
JSR	H_2 and N_2 species profiles	P: 1.2 atm T: 1000–1350 K Mixture: 8% NH_3 in Ar	[45]

*PW signifies present work.

**Fig. 9.** Comparison between the proposed mechanism (lines: NH , dashed lines: NH_2) and shock-tube experimental data from Davidson et al. [47] for NH_2/NH peak mole fraction in mixtures of 0.1–1% NH_3 in Ar.

els present in the mixture. Moreover, the sensitivity coefficients of the secondary reactions are reduced (compared with those of mixture 1) due to part of the generated NH_2 reacting with the existing H_2 via the forward direction of R5 and limiting the available NH_2 which influences most of the secondary reactions.

4.2. Model validation

To assess the present mechanism performance against other data, a literature survey was performed to collect experimental NH_3 pyrolysis data. Generally, there is a limited number of experimental studies targeted to investigate NH_3 pyrolysis. To the authors' knowledge, there are only four other studies where NH_3 pyrolysis kinetics data are reported. These studies along with their experimental details are summarized in Table 2.

Davidson et al. [47] performed shock-tube experiments to measure NH_2 and NH profiles during pyrolysis of 0.01–1% NH_3 in Ar. In their study, NH_2 and NH peak values as a function of temperature for five different mixtures were provided. Fig. 9 shows a comparison between the present numerical prediction and the Davidson et al. data. The present mechanism shows excellent agreement with the NH_2 peak data of both mixtures for the entire range of temperatures. Additionally, the present mechanism well predicts

the NH peak data for the low-temperature region (2200–2900 K), while a small over prediction of the peaks is observed at higher temperatures (>3000 K). Note that in the Davidson et al. study, only an average pressure was provided for all the experiments, which was used to perform the simulations herein.

Michel and Wagner [62] studied hydrazine pyrolysis in a shock tube for mixtures of 0.03–0.5% N_2H_4 in Ar. In their study, they reported N_2H_4 half-life times as a function of temperature. The prediction of the present mechanism is compared with the N_2H_4 half-life times for two different mixtures in Fig. 10. The present mechanism produces excellent agreement with the experimental data for both mixtures. It should be noted that sensitivity analysis shows that at temperatures higher than ~ 1300 K, the only sensitive reaction in determining the N_2H_4 half-life times is R37 ($\text{NH}_2 + \text{NH}_2 + \text{M} \rightleftharpoons \text{N}_2\text{H}_4 + \text{M}$). Multiple rate coefficients for R37 are reported in the literature [66,73,74,80], yet the rate coefficient provided by Klippenstein et al. [70] produced the most-accurate prediction of the experiments. At temperatures lower than 1300 K, R35 and R36 are important in accelerating the consumption of N_2H_4 , and removing these two reactions yielded ~ 50 – 80% longer N_2H_4 half-life times compared with the experimental data.

Recently, Benés et al. [63] measured NH_3 conversion and H_2 formation during NH_3 pyrolysis in a flow reactor. Their experiments were performed for $\sim 0.1 - 1.0\%$ NH_3 in both Ar and N_2 in two types of reactors. Figure 11 shows a comparison of the numerical prediction from the present mechanism with the results from Benés et al. for a mixture of 994 ppm NH_3 in Ar. The computations were performed using the Chemkin plug-flow reactor module by modeling the reactors with a fixed temperature, using the experimental wall temperature measurements provided by Benés and coworkers. The NH_3 conversion data are well predicted by the present mechanism for the entire range of temperatures, while the present mechanism predicted accurately the H_2 concentration for temperatures below ~ 1600 K. Above this temperature, the model slightly under predicts the H_2 formation, yet the predictions are very close ($\sim 5\%$ difference) considering the uncertainty of the measurements. It should be noted that modeling the flow reactor data by fixing the reactor temperature and using the residence time (2.3 – 4.0 s) provided by Benés et al. yielded similar predictions at temperatures below ~ 1500 K, but slightly under predicts (e.g. $\sim 10\%$ at 1700 K) both experimental data at higher temperatures.

Manna et al. [45] performed pyrolysis and oxidation of NH_3 in a Jet-Stirred reactor (JSR). In their pyrolysis experiments, they measured H_2 and N_2 concentrations for a reactor temperature of 1000–1300 K in a mixture of 8% NH_3 in Ar. The prediction

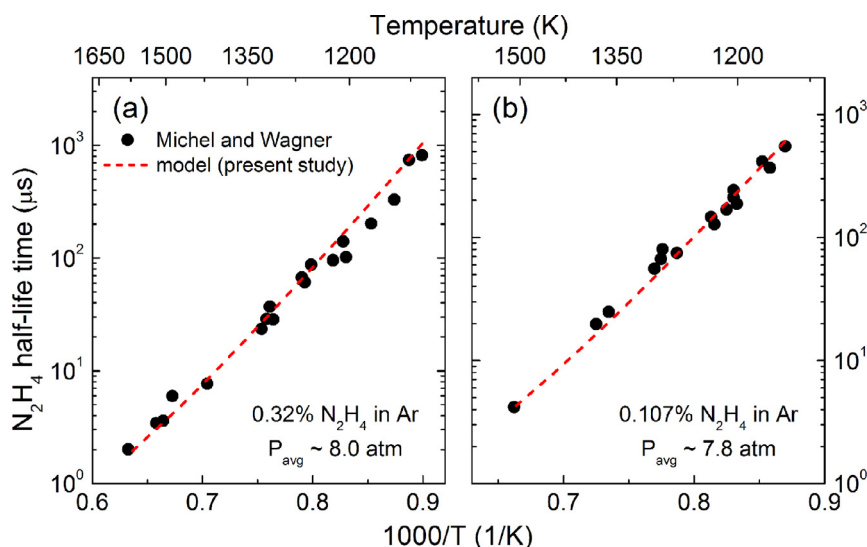


Fig. 10. Comparison between the proposed mechanism and shock-tube data from Michel and Wagner [62] for N_2H_4 half-life times in mixtures of (a) 0.32% N_2H_4 , and (b) 0.107% N_2H_4 in Ar.

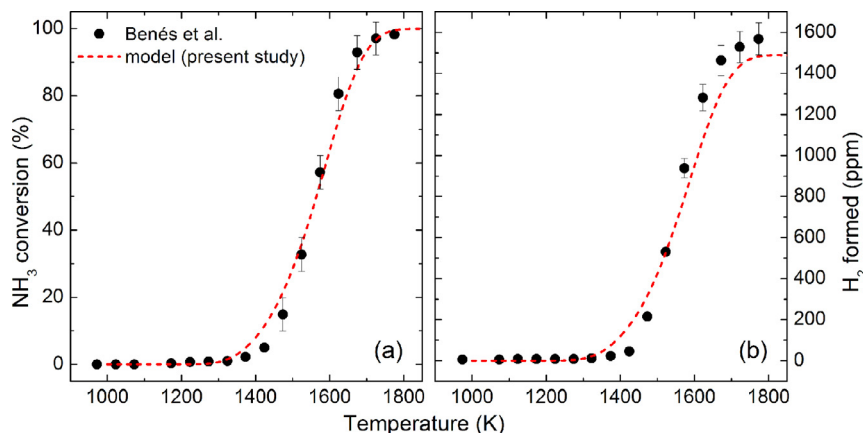


Fig. 11. Comparison between the proposed mechanism and flow reactor experimental data from Benés et al. [63] for (a) NH_3 conversion and (b) H_2 formed mole fraction in a mixture of 994 ppm NH_3 in Ar.

from the present mechanism of Manna et al. pyrolysis results are shown in Fig. 12. The modeling was performed via the PSR code in Chemkin using the reactor inlet temperature and the residence time of 0.25 s provided in the Manna et al. study. The present mechanism predicts no significant formation of H_2 and N_2 for the experimental range of temperatures up to ~ 1400 K as NH_3 decomposition is limited within the residence time. The present mechanism predicts that NH_3 decomposition starts at temperatures above 1500 K for the experimental conditions of Manna et al., which is followed by the formation of H_2 and N_2 as seen in Fig. 12. The numerical analysis showed that the most-sensitive reaction to the formation of both H_2 and N_2 is the main NH_3 decomposition reaction R4. It should be noted that attempts to model these JSR data using other mechanisms from the literature were performed, yet none of the models considered herein were able to predict any reactivity at the temperature range of the experiments (i.e. 1000–1300 K, also see Fig. 9 in Manna et al.).

Note also that the residence time of the flow reactor data by Benés et al. (shown in Fig. 11) was anywhere between 2.3 and 4.0 s, and the mixture reactivity was observed at temperatures above 1400 K, while the residence time of the mixture in the JSR data was 0.25 s and the reactivity started at a lower temperature of ~ 1100 K. Although the concentrations of NH_3 in the mix-

tures were different in the two studies, this indirect, yet conflicting, comparison between the two experiments coupled with the significant deviation of the prediction from the present model as well as other literature models may indicate some problems with these data, where the over-reactivity could be induced by surface reactions. Note that several changes were made to the mechanism proposed in the current study to model these data, but it was not possible to reconcile the predictions for the JSR data from Manna et al. and the other data considered herein. This discrepancy could also be indicative of a missing pathway in current models and new NH_3 pyrolysis measurements would be necessary to confirm the data from Manna and coworkers.

Overall, the new model well reproduced the present experiment data, along with other available NH_3 pyrolysis literature data. However, it should be noted that the experimental pyrolysis data used for the present model validation were all obtained near atmospheric pressure, except for the Michel and Wagner [62] hydrazine pyrolysis data. Note also that NH_3 pyrolysis experimental data do not exist at high pressure in the literature. Thus, the validation of the present mechanism at high pressure was not feasible at this time. Additionally, the selection of the kinetics parameters in Table 1 were all adopted as per the original source which did not include pressure dependency factors, and most of the NH_3 pyroly-

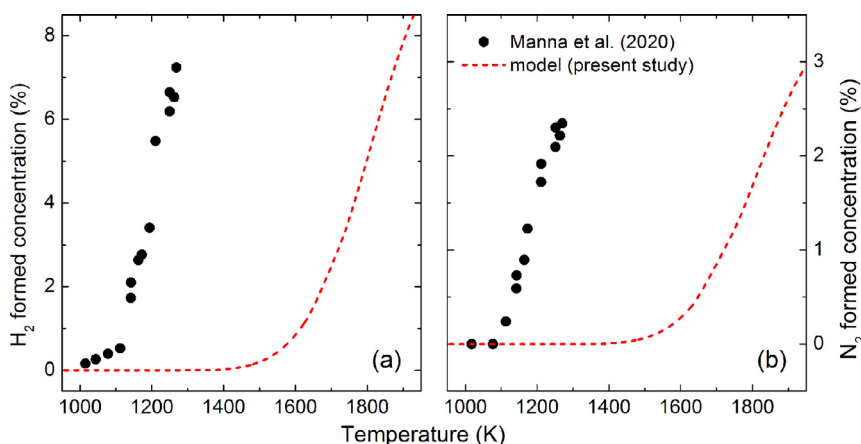


Fig. 12. Comparison between the proposed mechanism and JSR experimental data from Manna et al. [45] for (a) H_2 concentration and (b) N_2 concentration in a mixture of 8% NH_3 in Ar.

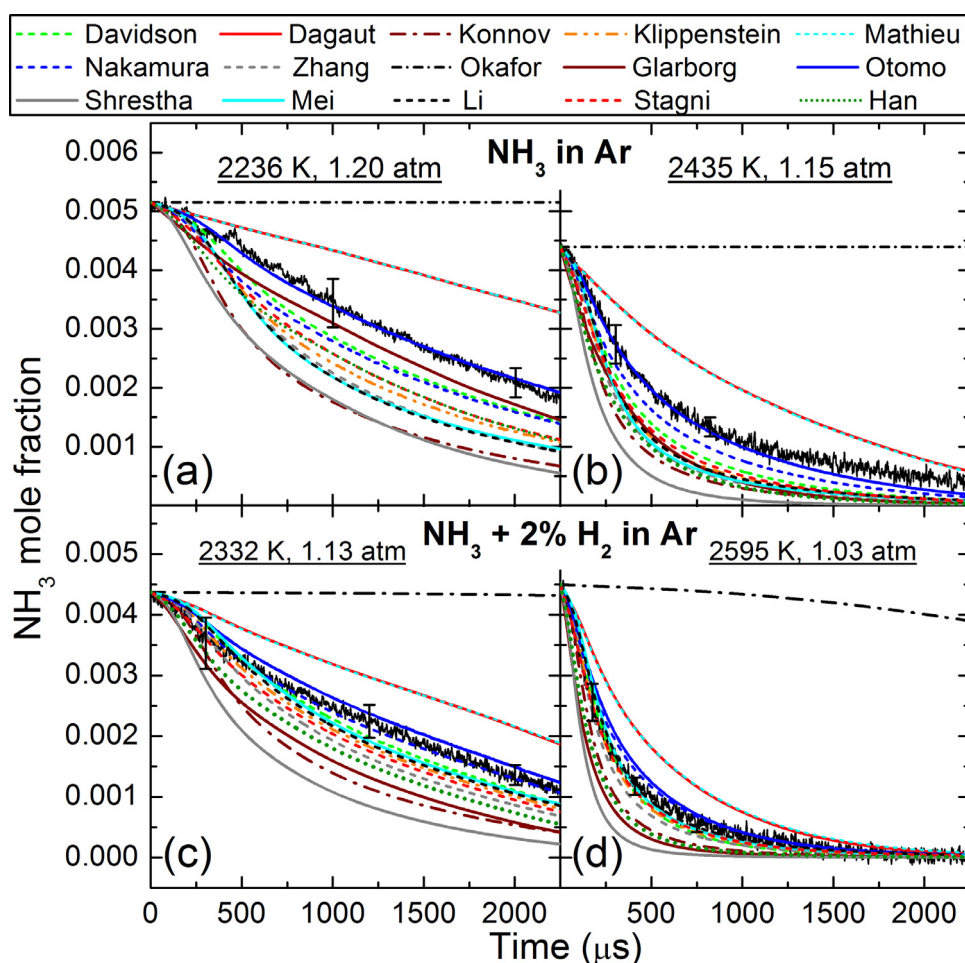


Fig. 13. Comparison of the experimental NH_3 speciation profiles and numerical predictions from available kinetics mechanisms for a mixture of (a) 0.515% NH_3 , (b) 0.440% NH_3 , (c) 0.437% $\text{NH}_3/2\% \text{H}_2$, (d) 0.450% $\text{NH}_3/2\% \text{H}_2$ balance Ar.

sis reactions simply do not have available kinetics parameters that include pressure dependency.

4.3. Performance of available NH_3 models

Recently, a large number of detailed chemical kinetics mechanisms of NH_3 were reported in the literature. The predictions from 14 detailed kinetics mechanisms of NH_3 published in the last 12 years were assessed to validate the model accuracies in reproduc-

ing the experimental data from the present study. Additionally, the Davidson et al. [47] pyrolysis mechanism was also included to highlight the improvement made in the present pyrolysis mechanism.

Figure 13 presents the predictions from the literature models of four selected NH_3 profiles for mixture 1 (Fig. 13a-b) and mixture 2 (Fig. 13c-d). The modeling was performed using the VITM method, as discussed in section 1.3, to accurately depict the experimental shock-tube conditions. For mixture 1 experiments, the majority

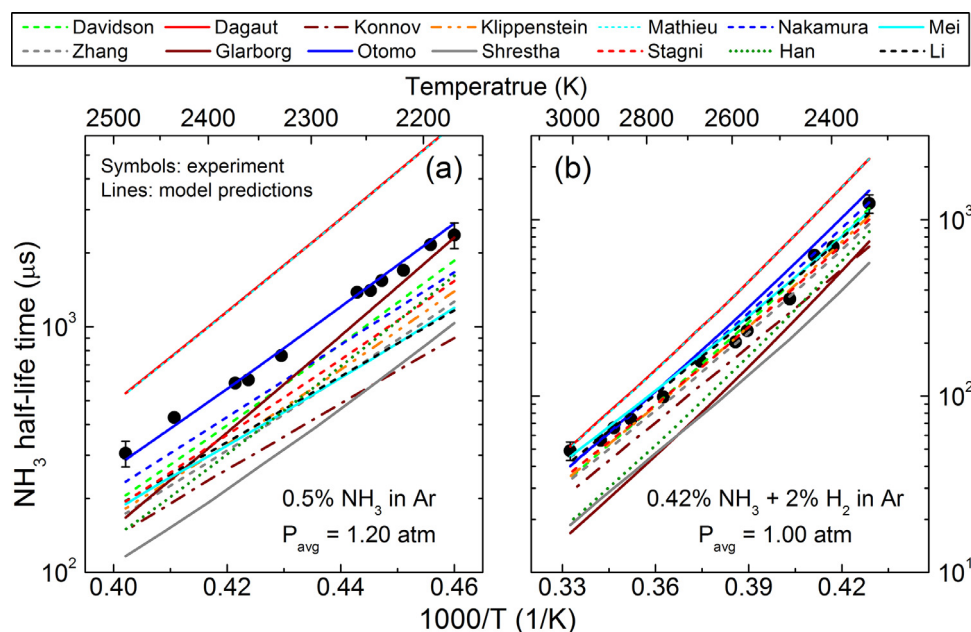


Fig. 14. Comparison of the experimental NH_3 half-life times and numerical predictions from literature kinetics mechanisms. (a) Experiments performed at 1.12–1.26 atm for a mixture of 0.41%–0.52% NH_3 in Ar. Modeling performed at an average pressure of 1.20 atm for 0.5% NH_3 in Ar. (b) Experiments performed at 0.88–1.13 atm for a mixture of 0.36%–0.45% $\text{NH}_3/2\%$ H_2 in Ar. Modeling performed at an average pressure of 1.00 atm for 0.42% $\text{NH}_3/2\%$ H_2 in Ar.

of the models (11 out of 15) produced a faster rate of NH_3 consumption to different degrees. From these 11 models, the models of Shrestha et al. [34] and Konnov [49] produced the furthest deviation (fastest rate of consumption) from the experiments, while the models of Nakamura et al. [40] and Davidson et al. [47] deviated the least. On the other hand, only two models, Mathieu and Petersen [5] and Dagaut et al. [48], produced a slower rate of NH_3 consumption. Analysis of these two models showed that this slower rate was due to the missing reaction R10 ($\text{NH}_2 + \text{NH} \rightleftharpoons \text{N}_2\text{H}_2 + \text{H}$) which ultimately eliminates N_2H_2 chemistry (since R10 is the major pathway to form N_2H_2) and significantly lowers the rate of NH_3 consumption. The model of Okafor et al. [51] predicts no change in the NH_3 mole fraction which is due to this model missing the NH_3 decomposition reaction R4 that initiates the reactivity for mixture 1. Finally, the model of Otomo et al. [52] produced excellent agreement with the experimental profiles for both temperature conditions. Note that the prediction of the Otomo et al. model is very similar to that of the present mechanism for the NH_3 speciation data herein, yet the present mechanism produced improved predictions of the other literature pyrolysis data presented previously.

The bottom part of Fig. 13 presents the comparison of the models with the selected experiments conducted for the $\text{NH}_3/\text{H}_2/\text{Ar}$ mixture. The majority of the models predict a faster NH_3 rate of consumption compared with the experiments. Similar to the experiments without H_2 , the models of Shrestha et al. [34] and Konnov [49] produced the fastest rate of consumption, and the models of Mathieu and Petersen [5] and Dagaut et al. [48] produced the slowest rate of consumption. Additionally, the models of Glarborg et al. [42] and Han et al. [54] predicted faster reactivity especially for the highest temperature condition (Fig. 13d). The model of Okafor et al. [51] predicts no change in the NH_3 mole fraction for the low-temperature point and a small reduction through the experiment for the high-temperature point. This small reduction is due to the small amount of H-atoms (generated from the H_2 decomposition via R3) reacting with NH_3 via -R5. Overall, several models produced good agreement with the

$\text{NH}_3/\text{H}_2/\text{Ar}$ experiments. The models of Nakamura et al. [40] and Otomo et al. [52] agree well with the experiments at the low- and high-temperature points, and the model predictions of Davidson et al. [47], Mei et al. [16], and Stagni et al. [43] showed closer agreement with the experiments for the high-temperature point.

To assess the performance of the models over the whole range of temperatures, the model predictions are compared with the NH_3 half-life times for both mixtures in Fig. 14. For mixture 1 (Fig. 14a), only the model of Otomo et al. [52] accurately reproduced the experimental measurements, while all other models failed in reproducing the experimental data with varying degrees of deviation. The models of Nakamura et al. [40] and Davidson et al. [47] showed the smallest deviation with $\sim 30\%$ faster NH_3 half-life times. All the models produced a similar slope for the NH_3 half-life time temperature dependency, except the models of Glarborg et al. [42] and Shrestha et al. [34], with the model of Glarborg et al. producing good agreement at low temperatures but diverging as the temperature increases. Note that the Okafor model was not considered here, due to its poor performance in Fig. 13.

The model predictions are compared with the NH_3 half-life times for mixture 2 in Fig. 14b. The majority of the models produced acceptable results including the models of Otomo et al. [52], Nakamura et al. [40], Mei et al. [16], Li et al. [53], Klippenstein et al. [50], and Zhang et al. [41]. The model of Konnov [49] was slightly too reactive but produced a similar slope to the experimental data, while the models of Glarborg et al. [42], Han et al. [54], and Shrestha et al. [34] were too reactive especially at higher temperatures. Note that the overall deviation of the models for mixture 2 is less due to the higher temperatures of the experiments compare with those of mixture 1. At high temperatures, the NH_3 half-life times are relatively short, and generally fewer reactions are involved in governing the NH_3 rate of consumption, while at lower temperatures an increasing number of important reactions affect the NH_3 profiles. These generally larger differences in model predictions at lower temperature for other parameters such as ignition delay time were also observed in other studies [5]. Overall, from the model comparison at all the experiment conditions

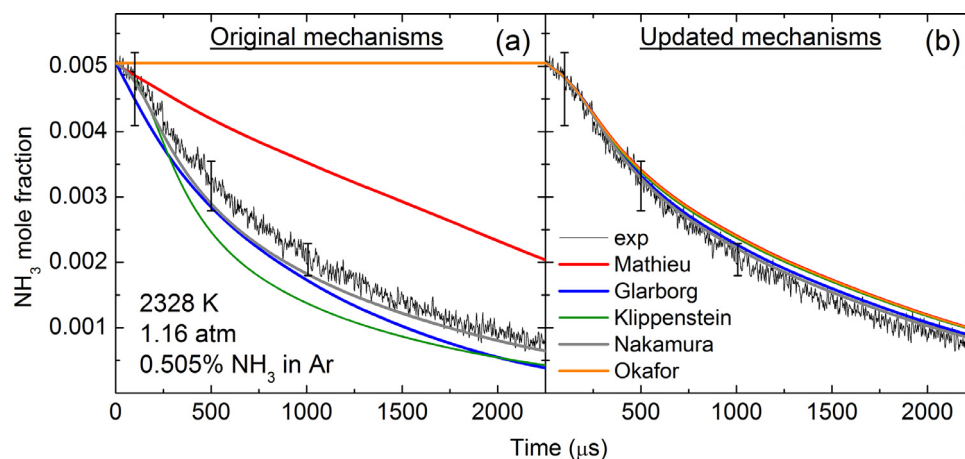


Fig. 15. Prediction of NH_3 speciation profile during pyrolysis showing the effect of updating selected detailed chemical kinetics mechanisms with the NH_3 pyrolysis sub-mechanism of the present study.

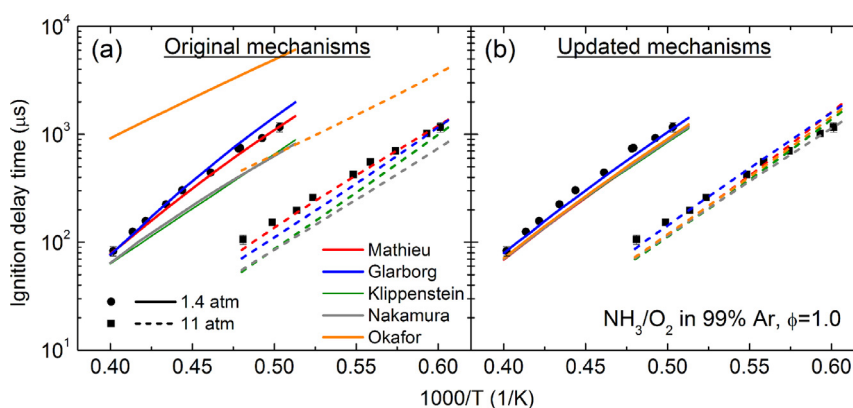


Fig. 16. Prediction of NH_3/O_2 ignition delay time data showing the effect of updating selected detailed chemical kinetics mechanisms with the NH_3 pyrolysis sub-mechanism of the present study. Experiments were taken from Mathieu and Petersen [5].

(Figs. 13 and 14), the model of Otomo et al. [52] is the only model that reproduced the NH_3 pyrolysis experiments to an acceptable degree.

It is of great interest to consider the effect of the NH_3 pyrolysis sub-mechanism on the performance of the detailed chemical kinetics models from the literature during both NH_3 pyrolysis and oxidation. Thus, the NH_3 pyrolysis sub-mechanism of selected literature models was updated with the pyrolysis mechanism proposed herein, for the sake of evaluating the impact of the pyrolysis sub-mechanism on predicting ammonia combustion chemistry. The literature models were updated by replacing the rate constants of the 48 reactions of the present mechanism with the selected rate constants presented in Table 1. Some of the literature models did not include the full list of reactions of the present mechanism, thus, the missing reactions were added. No alteration of any other reactions or the thermochemical data was performed.

Figure 15 shows the modeling results of a sample NH_3 speciation profile for the selected literature models (Fig. 15a) and the same models but updated with the NH_3 pyrolysis sub-mechanism from the present study (Fig. 15b). The updated models showed an improved prediction of the experimental profile. The small differences in the predictions could be attributed to the difference in the adopted thermochemistry of each study, in addition to the difference in H_2 chemistry in these models.

Additionally, the effects of updating the NH_3 pyrolysis sub-mechanism in the selected literature models were examined for

NH_3 oxidation. Mathieu and Petersen [5] reported ignition delay times for NH_3/O_2 in highly diluted Ar mixtures for a range of pressures and temperatures. Fig. 16 shows the ignition delay times of Mathieu and Petersen for stoichiometric NH_3/O_2 mixtures at two pressures of 1.4 and 11 atm. The ignition delay time predictions from the selected literature models are shown in Fig. 16a, while the prediction from the same models with the updated NH_3 pyrolysis sub-mechanism of the present study, the prediction improved for the models, especially the Okafor et al. [51] mechanism. Sensitivity analysis using the updated models confirmed that among the 10 most-sensitive reactions, 8 are NH_3 pyrolysis reactions, while the remaining two reactions are $\text{H} + \text{O}_2 \rightleftharpoons \text{O} + \text{OH}$ and $\text{NH} + \text{O} \rightleftharpoons \text{NO} + \text{H}$. Therefore, it can be concluded that NH_3 pyrolysis plays an important role in oxidation chemistry, at least for the high-temperature shock-tube conditions presented in Fig. 16. Generally, the improved performance of the updated models is due to important missing reactions and the selection of better kinetic parameters. For example, The Mathieu and Petersen and Okafor et al. mechanisms were missing the two important reactions of $\text{NH}_2 + \text{NH} \rightleftharpoons \text{N}_2\text{H}_2 + \text{H}$ and $\text{NH}_3 + \text{M} \rightleftharpoons \text{NH}_2 + \text{H} + \text{M}$, respectively, and the inclusion of these reactions improved the predictions. On the other hand, the remaining models included all the important reactions, and only the selection of better kinetic parameters improved the

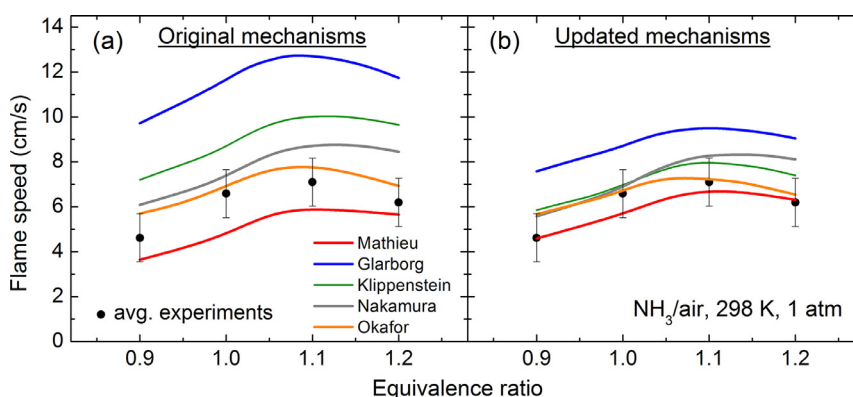


Fig. 17. Prediction of NH_3/air laminar flame speed showing the effect of updating selected detailed chemical kinetics mechanisms with the NH_3 pyrolysis sub-mechanism of the present study. Experiments were averaged at each equivalence ratio from the studies of Takizawa et al. [13], Hayakawa et al. [14], Mei et al. [16], Han et al. [28].

predictions. It should be noted that a full investigation of the impact of the pyrolysis sub-mechanism on NH_3 oxidation chemistry is beyond the scope of this work, and the previous illustration is to only highlight the importance of NH_3 pyrolysis sub-mechanism during oxidation.

It is also worth investigating the effect of the NH_3 pyrolysis sub-mechanism on the prediction of laminar flame speed. Figure 17 presents the laminar flame speed predictions from the selected literature models (solid lines), and from the updated models with the NH_3 pyrolysis sub-mechanism of the present study. Additionally, the average flame speed for each equivalence ratio from four previous experimental studies [13,14,16,28] are presented in Fig. 17. The presented uncertainty is taken as the standard deviation of the data and translates to $\sim \pm 1$ cm/s, which is relatively consistent with the uncertainties provided in the original flame speed studies. For all the selected models, the update in the NH_3 pyrolysis sub-mechanism improved the laminar flame speed prediction compared with the experimental results. The laminar flame speed predictions of the Mathieu and Petersen [5] increased $\sim 14\%$, while the predictions from the models of Klippenstein et al. [50], Okafor et al. [51], and Nakamura et al. [40] decreased approximately 14%, 7%, and 5%, respectively. On the other hand, the predictions from the model of Glarborg et al. [42] improved significantly with $\sim 25\%$ decrease in the flame speed predictions when updated with the NH_3 pyrolysis sub-mechanism of the present study. This improvement of prediction in laminar flame speed from the selected model further highlights the importance of NH_3 pyrolysis chemistry.

5. Conclusions

The chemical kinetics of ammonia during pyrolysis was studied experimentally by following NH_3 rate of consumption in a shock tube. A new quantitative laser absorption diagnostic was used to measure the mole fraction of NH_3 over a range of high-temperature conditions. Experiments to measure NH_3 speciation profiles were performed near atmospheric pressure for a temperature range of 2100–3000 K in two types of mixtures: $\sim 0.5\%$ NH_3 in Ar and ~ 0.42 $\text{NH}_3/2\%$ H_2 in Ar. The experimental results were used as a target to assemble a new NH_3 pyrolysis mechanism. The numerical prediction of the new mechanism showed good agreement with the NH_3 speciation profiles. The new mechanism was validated against other experimental data of NH_3 pyrolysis available in the literature. The numerical predictions of the new mechanism were in good agreement with the available literature data, except for one data set, and a discussion of possible reasons for disagreement was included. Additionally, the performance of sev-

eral detailed chemical kinetics mechanisms of NH_3 was examined against the experimental data of the present study. Overall, the chemical kinetics mechanisms from the literature failed to reproduce the experimental data, especially for the NH_3/Ar experiments. The exception was the model of Otomo et al. [52] which produced good overall agreement with the experimental data. Several detailed chemical kinetics models were used to investigate the effect of updating the NH_3 pyrolysis sub-mechanism to the results of the present mechanism. The effect of the NH_3 pyrolysis sub-mechanism was examined for both pyrolysis and oxidation, and it was shown that NH_3 pyrolysis plays an important role during high-temperature oxidation. Based on its performance against the limited set of data available for NH_3 pyrolysis, the model proposed in the present study could constitute a base to build upon to develop and validate future NH_3 mechanisms in more complex systems.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2021.111694.

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